

RESEARCH ARTICLE

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Key Points:

- We develop a statistical model to estimate the probability and spatial extent of seismically induced landslides for major earthquakes around the world within minutes after an earthquake occurs
- Earthquake-induced landslides are shown to be related to peak ground velocity, slope, lithology, land cover type, and ground wetness
- This model uses globally available data and can be applied in near-real-time to any earthquake around the globe for which a USGS *ShakeMap* is produced

Supporting Information:

- Supporting Information S1
- Table S1

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A Global Empirical Model for Near-Real-Time Assessment of Seismically Induced Landslides

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Abstract Earthquake-triggered landslides are a significant hazard in seismically active regions, but our ability to assess the hazard they pose in near-real-time is limited. In this study, we present a new globally applicable model for seismically induced landslides based on the most comprehensive global data set available; we use 23 landslide inventories that span a range of earthquake magnitudes and climatic and tectonic settings. We use logistic regression to relate the presence and distribution of earthquake-triggered landslides with spatially distributed estimates of ground shaking, topographic slope, lithology, land cover type, and a topographic index designed to estimate variability in soil wetness to provide an empirical model of landslide distribution. We tested over 100 combinations of independent predictor variables to find the best fitting model, using a diverse set of statistical tests. Blind validation tests show that the model accurately estimates the distribution of available landslide inventories. The results indicate that the model is reliable and stable, with high *balanced accuracy* (correctly versus incorrectly classified pixels) for the majority of test events. A cross-validation analysis shows high balanced accuracy for a majority of events as well. By combining near-real-time estimates of ground shaking with globally available landslide susceptibility data, this model provides a tool to estimate the distribution of coseismic landslide hazard within minutes of the occurrence of any earthquake worldwide for which a U.S. Geological Survey *ShakeMap* is available.

1. Introduction

As a natural component of erosive processes, landslides present hazard in areas with large topographic relief and slope and can result in significant loss of human life and damage to the built environment (Daniell et al., 2017; Marano et al., 2009). Although landslides can be triggered by earthquakes, precipitation, and by human activities, here we focus on landsliding induced by earthquakes. Marano et al. (2009) showed that 5% of fatalities related to earthquakes were caused by landsliding and that they were the third largest contributor to fatalities after building collapse and tsunamis. In two recent major earthquakes in Wenchuan, China (2008), and Nepal (2015), there was significant impact due to landslide triggering; the Wenchuan, China, event alone was estimated to have triggered 197,481 landslides (Xu, Xu, Yao, et al., 2014), and landslides triggered by these earthquakes caused an estimated 20,120 fatalities (Petley, 2015; Yin et al., 2009). Earthquake-induced landslides (EQILs) have also been shown to be a major cause of disruption to lifelines in mountainous regions (Bird & Bommer, 2004), impeding emergency response efforts. Damage to transportation lifelines can impede emergency responder access to affected areas, causing delays of search and rescue efforts as well as delivery of aid.

Producing estimates to rapidly assess the extent and distribution of hazard due to seismically induced landsliding in near real time is thus an important aim, as this information could be used in postseismic response efforts to identify populations and lifelines likely impacted by landsliding. Estimates of the overall extent of landslide hazard can also be helpful for evaluating response strategies immediately following the occurrence of an earthquake. Until recently, few studies have addressed real-time assessment of EQIL. Currently, four classes of published models could be applied regionally or globally if the required inputs are available: physical mechanistic models based on the method of Newmark displacement (e.g., Jibson et al., 2000; Godt et al., 2008; Gallen et al., 2017), statistical models developed with logistic regression using landslide inventories (e.g., Nowicki et al., 2014; Parker et al., 2017), statistical models based on fuzzy logic (e.g., Kritikos et al., 2015; Robinson et al., 2018), and statistical models that relate the total area and

volume of landsliding to seismologic parameters of individual earthquakes (Marc et al., 2016). Our previous model (Nowicki et al., 2014) was the first statistical model that could be readily applied globally for near-real-time prediction of EQIL.

In this article, we present a significant expansion of the observational data set and the methodology used in our previous global model, incorporating significantly more data and analyses of the model's performance in a near-real-time setting. The Nowicki et al. (2014) study used four well-documented observational inventories. Since then, we have expanded the observational landslide data set with 18 additional events from a wider variety of geologic and geographic environments, from which we develop a landslide model by combining landslide susceptibility proxies (including topographic slope, surface geology, wetness, and high-resolution land cover data) with shaking estimates (derived from the U.S. Geological Survey (USGS) *ShakeMap* system; Worden & Wald, 2016). The model estimates the probability (i.e., the relative hazard in comparison to other areas of the map) of landsliding triggered by individual earthquakes by combining near-real-time estimates of ground shaking with globally available landslide susceptibility data. This model provides a tool to predict seismically induced landslides across the globe within minutes after substantial earthquakes worldwide in a systematic way.

2. Data

Behavior, location, size, and mobility of earthquake-triggered landslides are all controlled by physical properties and hydrologic conditions of near-surface materials and the characteristics of ground shaking during an earthquake. Empirical studies suggest that the bedrock lithology, slope, seismic intensity, topographic amplification of ground motion, fracture systems in the underlying bedrock, groundwater conditions, and also the distribution of preexisting landslides all have some impact on landslide distribution. These factors combine to affect the static and dynamic shear stress of a hillslope as well as the material strength resisting downslope movement (e.g., Keefer, 2002). Consistent maps of most of those factors rarely exist regionally, let alone globally. We use these studies that describe known impacts on the distribution of landsliding to guide our selection of globally available predictor variables that can represent the susceptibility of an area to landsliding. Based on these guidelines, we initially include the following predictor variables (for which global datasets are available in the regression analysis) for further testing: (1) ground motions produced by the earthquake, (2) topographic slope, (3) elevation, (4) lithology, (5) soil wetness, (6) precipitation, (7) land cover, and (8) earthquake magnitude.

2.1. Landslide Data (Dependent Variable)

Central to our analysis are the location and size of mapped landslides triggered by individual earthquakes. Currently, we have access to landslide data from 36 EQIL inventories, a subset of those summarized by Tanyas et al. (2017; see Table 1 and Figure 1). Twenty-one of these inventories are now publicly available in Schmitt et al. (2017). This data set includes landslides mapped by a variety of methods including field-based mapping of observed landslide deposits, and remote sensing techniques. There are two common methods of reporting the location of mapped landslides: (1) each landslide is mapped as an area (with polygons that rarely denote the source and slide runout areas separately) or (2) each landslide is mapped as a single point (this varies with each author and could potentially denote the centroid of each landslide, the source of the landslide, or any point within the slide). Table 1 represents all of the EQIL inventories that we include in this analysis. These data include landslides mapped as point and polygon data and also represent data of variable quality, as described below.

In order to use a majority of these inventories while reducing the amount of low-quality data in our global dataset, we follow the scoring technique described by Tanyas et al. (2017). For this model, we utilize their *essential criteria* required for inventories to be used for hazard assessment; these include whether (1) the study area was systematically analyzed by visual interpretation using satellite imagery (as compared to landslides mapped using automatic image classification methods; 0–100% value), (2) a boundary of the mapping area was indicated (no/yes as 0 or 1), (3) landslides that occurred before and after the earthquake were removed from the inventory (0–100% value), and (4) the landslides were mapped at a high enough resolution to recognize individual landslides (0–1 value based on the smallest landslides that were mapped in that inventory). We convert the percentage scores to a 0–1 value and add these four values together to create an overall score for these four essential criteria. We exclude any inventories that scored less than a

Table 1
Summary of Landslide Inventories Currently Available

Event (location)	Date	Year	M_w	Landslide data type	Number of landslide observations	Data quality score	Reference
San Fernando, California	9 February	1971	6.6	Points	391	0.8	Morton (1971)
Guatemala	4 February	1976	7.5	Polygons	6224	3.5	Harp et al. (1981)
Friuli, Italy	5 May	1976	6.5	Points	1007	2.3	Govi (1977)
Coalinga, California	2 May	1983	6.7	Polygons	3980	3.5	Harp and Keefer (1990)
San Salvador, El Salvador	10 October	1986	5.7	Points	268	1.3	Rymer (1987)
Loma Prieta, California	17 October	1989	6.9	Points	528	1.5	Keefer and Manson (1998)
Northridge, California	17 January	1994	6.7	Polygons	11111	3.1	Harp and Jibson (1996)
Kobe, Japan	16 January	1995	6.9	Polygons	2353	3.6	Uchida et al. (2004)
Chi-Chi, Taiwan	20 September	1999	7.7	Polygons	9272	2.4	Liao and Lee (2000)
El Salvador	13 January	2001	7.7	Points	139	1.5	Ministerio de Medio Ambiente y Recursos Naturales (2017)
El Salvador	13 February	2001	6.6	Points	62	1.5	Ministerio de Medio Ambiente y Recursos Naturales, El Salvador (2017)
Avaj, Iran	22 June	2002	6.5	Points	50	1.2	Mahdavifar et al. (2006)
Denali, Alaska	3 November	2002	7.9	Polygons	1579	2.6	Gorum et al. (2014)
Lefkada, Greece	14 August	2003	6.3	Polygons	274	2.3	Papathanassiou et al. (2013)
Niigata-Chuetsu, Japan	23 October	2004	6.6	Polygons	4615	3.5	Sekiguchi and Sato (2006)
Kashmir, Pakistan	8 October	2005	7.6	Points	2424	3.5	Sato et al. (2007)
Kiholo Bay, Hawaii	15 October	2006	6.7	Polygons	383	3	Harp et al. (2014)
Niigata-Chuetsu-oki, Japan	16 July	2007	6.6	Points	312	2.4	Kokusai Kogyo (2007)
Aysen Fjord, Chile	21 April	2007	6.2	Polygons	517	3.3	Gorum et al. (2014)
Wenchuan, China	12 May	2008	7.9	Polygons	197481	3.4	Xu, Xu, Yao, et al. (2014)
Iwate-Miyagi-Nairuku, Japan	14 June	2008	6.9	Polygons	4211	3.6	Yagi et al. (2009)
Abruzzo, Italy	6 April	2009	6.3	Polygons	89	3	Piacentini et al. (2013)
Sumatra, Indonesia	30 September	2009	7.6	Points	87	1	Umar et al. (2014)
Haiti	12 January	2010	7	Polygons	23567	3.8	Harp et al. (2016)
Sierra Cucapah, Mexico	4 April	2010	7.2	Polygons	453	3.6	Barlow et al. (2015)
Yushu, China	14 April	2010	6.9	Polygons	2036	3.8	Xu et al. (2013)
Lorca, Spain	11 May	2011	5.1	Points	166	0.6	Alfaro et al. (2012)
Tohoku, Japan	11 March	2011	9.1	Polygons	3477	3.4	Wartman et al. (2013)
Lushan, China	20 April	2013	6.6	Polygons	15546	3.8	Xu et al. (2015)
Cook Strait, N.Z.	21 July	2013	6.5	Points	35	0.5	Van Dissen et al. (2013)
Minxian-Zhangxian, China	21 July	2013	5.9	Polygons	2330	3.8	Xu, Xu, Shyu, et al. (2014)
Lake Grassmere, N.Z.	16 August	2013	6.5	Points	501	0.6	Van Dissen et al. (2013)
Eketahuna, N.Z.	20 January	2014	6.1	Points	176	0.5	Rosser et al. (2014)
Ludian, China	3 August	2014	6.2	Polygons	1024	3.4	Ying-Ying et al. (2015)
Wilberforce, N.Z.	6 January	2015	5.6	Points	265	0.5	GNS Science (2015)
Kumamoto, Japan	15 April	2016	7	Polygons	336	3.5	DSPR-KU (2016)

Note. Those with a data quality score of 2.0 or above are used for training purposes in the global landslide model (shown in bold). All others were used for testing the model. Due to the small number of landslide observations available, the Abruzzo, Italy inventory was used for testing purposes.

value of 2, indicating that they meet less than half of these criteria. We use these inventories later for blind validation analysis. Of the 36 landslide inventories available at the time of this study, 13 were excluded on the basis of these criteria. We then combine the remaining 23 inventories into a single, global, database of landslide observations to be used for training the global landslide model.

2.1.1. Sampling Method for Landslide Data

We aimed to treat point and polygon data in such a way that they can both be incorporated into the landslide model accurately. This is because there are relatively few data sets of nearly complete (e.g., mapping the entire area in which landslides may have occurred, as compared to investigating portions of the region), high-quality polygon data, and even polygon data are represented differently in various studies. To maximize the number of data sets we can include, we apply a sampling scheme based on the method used by Zhu et al. (2017) for liquefaction observations and Den Eeckhaut et al. (2010) for landslide observations. This sampling scheme aims to produce an equally balanced set of landslide and nonlandslide observations, while allowing for landslides mapped as both polygons and points to be accurately incorporated into the database.



Figure 1. Map of events incorporated into two phases of modeling: 1. Training (gray symbols) and 2. Testing (white symbols) of the global database. Polygon data are shown as diamonds; point data are shown as dots.

Landslides mapped as polygons constrain both landslide and nonlandslide occurrences. Landslides mapped as points, however, only directly indicate positive landslide occurrences. In order to overcome this fundamental limitation in the point data, we utilize a two-buffer algorithm, as described by Zhu et al. (2017). An inner buffer acts as a circular boundary around the landslide point, simulating a landslide polygon and excluding this area from being considered as a *nonlandslide* sample. Because there are no observations of nonlandslide locations, an outer (circular) buffer around the landslide point is used, such that any point that lies between the inner and outer buffers can be sampled as a nonlandslide observation (this method is similar to that of Den Eeckhaut et al., 2010). We randomly sampled nonlandslide points from within these two buffers. This assumes that nonlandslide observations are close enough to the *landslide* points, so that they are likely to have been examined in the survey when the points were mapped, and also provides an automated method to define the likely limit of the mapped extent of the inventory. The buffer sizes are defined empirically based on the mapped landslide polygons. The inner buffer is defined as the 95th percentile of the average radii of the polygon training inventories, providing a value for the average landslide size in the training data. We chose this inner buffer size to ensure that areas that are close enough to mapped landslide points to have possibly included landslides are not inadvertently sampled as nonlandslide observations; this for potential variation between landslide inventories, since the average radii calculation assumes perfectly circular polygons. The outer buffer is defined as the 95th percentile of the distance between the centroids of the polygons. We use this distance as a proxy for areas most likely to have been mapped surrounding each landslide indicating that this is a likely area of nonlandslide occurrence. For this data set, the algorithm allows sampling of nonlandslide sites between an inner buffer of 85 m and an outer buffer of 385 m from known landslide sites (see Figure S1 in the supporting information). Testing of a range of buffer radii revealed that the regression model is not sensitive to changes in these values, and this choice does not significantly impact the regression model.

In order to equally train the model toward positive and negative observations, we balanced the sample by randomly sampling a number of nonlandslide observations that is equal to the number of landslide observations in each individual earthquake, resulting in 50% landslide and 50% nonlandslide observations chosen from all of our available landslide data. These sampled observations from individual earthquakes are then combined to form the data set we use to fit the model. This method balances the ability of the model to represent landslide and nonlandslide observations. The modeled probabilities will then be inflated as compared to using an input data set containing the entire map of landslide and nonlandslide areas, which would have a very low percentage of landslide observations. As discussed below, this bias should be considered when interpreting the modeled probabilities. We address this by developing a correction for this bias, as described in the analysis section. We recognize that different types of landslides behave differently and interact with hillslopes in different ways. At this stage, we do not differentiate between different types

Table 2
Independent Variables Evaluated for Use in the Logistic Regression Model

Factor	Variable representation	Data source	Resolution
Shaking	Peak Ground Acceleration (PGA)	USGS <i>ShakeMap</i> System (Worden & Wald, 2016)	1 km ²
	Peak Ground Velocity (PGV)	USGS <i>ShakeMap</i> System (Worden & Wald, 2016)	1 km ²
	Modified Mercalli Intensity (MMI)	USGS <i>ShakeMap</i> System (Worden & Wald, 2016)	1 km ²
Slope	Slope calculated from GMTED median elevation data	Danielson and Gesch (2011)	7.5c (~250 m ²)
	Maximum Slope	Verdin et al. (2007)	30c (~1 km ²)
Lithology	GLiM Global Lithology Data	Hartmann and Moosdorf (2012)	Vector Data
	Friction Angle	Godt et al. (2008)	30c (~1 km ²)
	Inferred Strength	Hartmann and Moosdorf (2012) and Nadim et al. (2006)	Vector Data
Elevation	Mean Elevation	Danielson and Gesch (2011)	7.5c (~250 m ²)
	Median Elevation	Danielson and Gesch (2011)	7.5c (~250 m ²)
Soil Wetness	Compound Topographic Index	Moore et al. (1991)	30c (~1 km ²)
Temporal Wetness	WorldClim mean monthly precipitation	Hijmans et al. (2005)	30c (~1 km ²)
	CGIAR-CSI Global Aridity	Trabucco and Zomer (2009)	30c (~1 km ²)
	CGIAR-CSI Global-Monthly Potential	Trabucco and Zomer (2009)	30c (~1 km ²)
	Evapotranspiration (PET)		30c (~1 km ²)
Vegetation Cover (Land Cover)	Globcover 2009 Land Cover	Arino et al. (2012)	300 m
	MODIS Land Cover	Broxton, Zeng, Sulla-Menashe, and Troch (2014)	15c (~500 m ²)
	Global Land Cover Share 2014	Latham et al. (2014)	30c (~1 km ²)
	Percent Green Vegetation Cover	Broxton, Zeng, Scheftic, and Troch (2014)	30c (~1 km ²)
Earthquake Magnitude/Duration	Moment Magnitude	USGS	—

Note. Bold indicates the variables chosen for use in the final model. USGS = U.S. Geological Survey.

of landslides; thus, at present, a rockfall is treated identically to a shallow soil landslide. We recognize that this is a limitation of the current data; separate regressions of subsets of landslide data based on classified data sets could be an area of future refinement of the model.

2.2. Independent Variables

We test a number of representations for each factor included in the model as an indicator of landslide susceptibility. Table 2 shows a summary of these predictor variables, with the chosen data representation in the final model shown in bold.

Ground motion. Many studies have shown that landslide patterns reflect properties of the triggering ground shaking (e.g., Keefer, 1984, 2002; Meunier et al., 2007; Nowicki et al., 2014; Rodriguez et al., 1999). Here we test a number of the final estimates of ground motion parameters available from the USGS *ShakeMap* system (Worden & Wald, 2016), including peak ground acceleration (PGA), peak ground velocity (PGV), and Modified Mercalli Intensity (MMI). We note, however, that the *ShakeMap* estimates of ground motion amplitudes evolve in the hours and days following an earthquake as new source models and strong-motion and intensity data become available. As a result, the landslide probabilities also change substantially, as shown by Allstadt, Jibson, et al. (2018). Note that *ShakeMap* estimates have variable uncertainty (Worden & Wald, 2016), an additional complexity not addressed in the current analysis.

Topographic Slope. Slope steepness affects slope stability (e.g., Budimir et al., 2015). We recognize that different types of landslides (e.g., rockfalls, shallow landslides, deep-seated slides) have different relationships to slope. A limitation of our method at this time is that we do not distinguish between different types of landslides. This is because most inventories are not classified by event type. Elevation data are available in a number of different formats and levels of spatial resolution, so we attempt to incorporate slopes computed from the highest resolution elevation data that are consistently available globally. We tested maximum slope values from Verdin et al. (2007). They computed slope from the 90-m Shuttle Radar Topography Mission (SRTM) elevation data, and downsampled by choosing the steepest of these 90-m slope values within a 30-arc sec (~1-km) grid (as chosen by Nowicki et al., 2014). We also test and ultimately choose to use an updated slope data set at 7.5-arc sec (~250-m) resolution, which is derived from the median elevation value from the Global Multi-resolution Terrain Elevation Data (GMTED) topography data (Danielson & Gesch, 2011). The GMTED elevation data combine many elevation data sets from around the globe to provide an accurate estimate of global elevations. These include void-filled SRTM data at 1-arc sec resolution (covering 80% of

land surfaces), along with various other data sets to fill in geographic areas outside of the SRTM and down-sampled to produce a regular grid at 7.5-arc sec resolution. While these data still provide a coarse representation of slope values, the GMTED elevation data set is the highest-resolution data set that is readily available for slope calculation globally. The slope values calculated from the GMTED elevation data represent slope values based on median elevations.

Lithology. Whether or not a slope fails, as well as how it fails, during an earthquake depends on the strength of the material that is shaken (Keefer, 1984). We have tested three representations of global data that can be used to represent material strength of bedrock in the global landslide model. The first proxy we tested is the global material strength data set presented in Godt et al. (2008), based in turn on a global geologic map (Hearn et al., 2003) and inferred cohesion values and friction angles of common geological materials (Selby, 1993). The cohesion and friction values can be used to represent the cohesive component and frictional component of material strength, respectively (e.g., Jibson et al., 2000). In this study, cohesion and friction are taken as proxy for the spatial variation of material strength, so we test only one of the two representations. The second proxy we test is a newly available, GLiM (global lithological map) data set available for the entire globe (Hartmann & Moosdorf, 2012). This data set provides the most detailed representation of lithology tested here by combining 92 regional lithological maps at the highest resolution available from across the globe. Here we test their 13 top-level classes of lithology (removing water bodies from the analysis) and also classify the lithologies into relative strengths based on the ranking system of Nadim et al. (2006). We then gridded the data at 250-m resolution (comparable to the topographic slope data). The details provided by these data are limited by the scale of each map but represent the highest-resolution global lithology map available. See Hartmann and Moosdorf (2012) for further information on how the data were compiled. Ultimately, we utilize this data set in the model. We recognize that soil cover is highly variable and plays a role in near-surface landsliding, which is not accurately reflected in the GLiM data.

Material Wetness. It is widely recognized that rainfall, as well as accumulation of water in the near surface, can affect the stability of a hillslope by changing the weight and gravitational potential energy of the material (Ray et al., 2010). To incorporate this variability into the model, we test the compound topographic index (CTI) as a proxy for potential soil wetness (Moore et al., 1991). CTI combines the slope value (α) and the contributing basin area (A) to estimate the spatial variability of wetness within a landscape,

$$CTI = \ln \left(\frac{A}{\tan(\alpha)} \right). \quad (1)$$

High CTI values thus result from lower slope values with larger drainage areas, while low CTI values result from higher slope values with smaller drainage areas. This value does not consider soil moisture directly but is dependent on the potential influence of topography on soil wetness (i.e., drainage area and proximity to a stream). We note that the CTI data tested here are at a coarser scale than the topographic slope data and may smooth over smaller-scale variability in the potential soil wetness value.

Temporal Wetness. To account for seasonal variation in wetness, we consider an estimate of mean monthly precipitation from the WorldClim database (Hijmans et al., 2005), which is based on average climatic data from 1950 to 2000. These data provide an estimate of how much precipitation an area is likely to receive during a given month of the year but do not represent the amount of actual precipitation received at any location. We also test two data sets that can be used as proxies for temporally varying soil wetness, from the CGIAR-CSI Global-Aridity and Global-PET Geospatial Database (Trabucco & Zomer, 2009). Both of these data sets are based on the WorldClim database, where the potential evapotranspiration (PET) value estimates the ability of the atmosphere to remove water through evapotranspiration (based on mean temperature and radiation at the top of the atmosphere), and the aridity index represents the ratio of the amount of precipitation available relative to that demanded by the atmosphere through evapotranspiration (higher value for more humid conditions, lower value for more arid conditions). Thus, both indices act as a proxy for the amount of moisture available in the atmosphere; the PET value represents temporal changes monthly, while the aridity index represents a time-limited average from the period 1950–2000.

Land Cover. Vegetation type and coverage can affect the composite strength of the soil-vegetation root matrix, which affects the stability of a slope. The overall effect on strength can be variable and highly dependent on localized properties. We therefore test land cover data in our model as a proxy for

vegetation cover. We consider four representations of global land cover: (1) Moderate Resolution Imaging Spectroradiometer (MODIS) satellite-based land cover data (Broxton, Zeng, Sulla-Menashe, & Troch, 2014), given at 15-arc sec resolution (~500 m) and based on data collected during 2001–2010 which were separated into 16 classes, where each class represents a type of land cover; (2) Global Land Cover Share data, based on multiple sources collected during 2014, given at 30-arc second (~1-km) resolution and separated into 10 classes (Latham et al., 2014); (3) Globcover 2009 data, available at 300-m resolution (the highest available resolution of the land cover data sets), based on satellite imagery from multiple sources during 2009 (Arino et al., 2012) and separated into 20 classes; and (4) a data set that represents the maximum green vegetation fraction (0–100%) in an area and is based on MODIS-derived normalized difference vegetation index data from 2001 to 2012, available at 30 arc sec (~1-km) resolution (Broxton, Zeng, Scheftic, & Troch, 2014).

Earthquake Magnitude. The magnitude of an earthquake may affect the likelihood of landsliding, as shown by Keefer (1984, 2002) and Rodriguez et al. (1999), as it is often considered a proxy for shaking duration in ground failure analyses (e.g., Federal Emergency Management Agency, 2003). Because of this, we test the moment magnitude as a predictor variable to increase or reduce the probability of landsliding due to individual earthquakes. We note that although the amplitude of the ground motion is largely already incorporated through the use of peak amplitude factors (e.g., PGA), larger magnitude earthquakes should have greater durations than smaller magnitude earthquakes even if the amplitudes are the same. For this reason, we include a measure of magnitude in addition to the peak amplitude estimates.

2.3. Data Processing

We resample the predictor variables to match the *ShakeMap* extent for each earthquake. These grids are then transformed to an orthographic projection specific to the midpoint of each grid. These projected grids are interpolated to 50-m spacing to ensure accurate representation of mapped landslide distributions, using either a bilinear or nearest neighbor method (dependent on data type; nearest neighbor was used for nominal data). These grids were then sampled with the balanced sampling method described previously.

3. Methods

3.1. Logistic Regression

Here we apply a logistic regression model, which is appropriate for modeling binary-dependent variables. As summarized by Budimir et al. (2015), this method is widely accepted and utilized in landslide susceptibility modeling. Logistic regression fits the observed outcome of landslide occurrence (mapped as a 0 or 1) to the logistic function. The logistic function represents the log of the odds of landslide occurrence, which can be expressed as a linear combination of the individual predictor variables included in the equation

$$\text{Logit}(P) = \ln\left(\frac{P}{1-P}\right) = a + bx_1 + cx_2 + dx_3 + \dots, \quad (2)$$

where P is the probability of landsliding, $x_1, x_2, x_3, \dots$ are explanatory variables, and $a, b, c, d, \dots$ are regression coefficients. The probability of landslide occurrence is

$$P(t) = \frac{1}{1 + e^{-t}}, \quad (3)$$

where $t = a + bx_1 + cx_2 + dx_3 + \dots$. The coefficients are estimated with maximum likelihood estimation (MLE) with the logistic method option in the generalized linear model function available in R (R Core Team, 2017).

3.2. Model Development Strategy

3.2.1. Analysis of Relationships Between Landslide Data and Independent Variables

In order to test the potential contribution of individual predictor variables to estimate the occurrence of landslides, we perform a correlation analysis between the individual predictor variables and landslide data prior to model testing. The standard correlation coefficient (i.e., Pearson's correlation coefficient) requires both variables to be continuous, so instead, we must use a sequence of procedures to determine the strength of the relationship between the binary landslide observations and the independent landslide susceptibility variables. To provide a first-order assessment of this relationship, we examine histograms of each predictor

stratified by landslide or nonlandslide observations. We also evaluate the bivariate logistic regression for each independent variable (e.g., red lines in Figures 2 and S2 after analysis shown in Zhu et al., 2017). Model coefficients are shown in Table S1. From that analysis, we interpret the relationship of landslide occurrence to that single variable (see Figures 2 and S2) and identify whether there is preliminary evidence of any correlation with that parameter. Certain variables show much stronger correlation with positive landslide occurrence than others; for example, high values of ground shaking (PGV and PGA) and slope both show strong relationships with landslide occurrence. In contrast, the data used to represent elevation, aridity, average precipitation, PET, aridity index, and earthquake magnitude show little to no relationship with mapped landslide locations.

As a more quantitative assessment, we also use the point biserial correlation coefficient (r_{pb}), which is appropriate for assessing relationships between continuous and dichotomous data:

$$r_{pb} = \frac{M_1 - M_0}{s_n} \sqrt{\frac{n_1 n_0}{n^2}}, \quad (4)$$

where M_1 and M_0 are the mean values of the continuous variable present for all values in group 1 and group 0 (landslide and nonlandslide), respectively; n_1 and n_0 are the number of observations in each group; and n is the total number of observations in the data set. Lastly, s_n is the pooled standard deviation of all data values. This is computed by calculating the standard deviation of group 1 and group 0 individually, then computing a weighted average of these values to reflect the number of observations in each group. This correlation value ranges from -1 to 1 , with a value of 0 indicating no correlation, a value of -1 indicating perfect anticorrelation, and a value of 1 indicating perfect correlation (Louangrath, 2014).

Because two of our landslide susceptibility data sets (lithology and land cover) comprise nominal data, we also calculate Cramer's V to determine the relationship between these nominal data and dichotomous data of the landslide observations. Cramer's V represents the strength of a relationship between two variables by calculating the chi-square statistic and correcting for the effects of sampling size:

$$V = \sqrt{\frac{\chi^2}{M \min\{(r-1), (c-1)\}}}, \quad (5)$$

where χ^2 is the chi-square value between the two variables, M is the total number of observations, r is the number of values in one of the ordinal datasets, and c is the number of values in the other. This value ranges from 0 to 1 ; as the value approaches 1 this indicates a stronger relationship between the variables (Reynolds, 1984).

Computing correlation coefficients between these individual predictor variables and landslide occurrence (Figure 3) supports the qualitative interpretation discussed previously with a quantitative value. If these methods show that no statistically significant relationship exists (e.g., r_{pb} or $V < 0.1$, indicating no or a very weak relationship) between landslide occurrence and a particular variable, we remove the variable from the model testing process and move forward with the remaining predictor variables to test within our model framework.

3.3. Model Testing and Selection

To select between different possible combinations of model functional forms, we compute the following:

1. the Akaike Information Criteria (AIC; Wagenmakers & Farrell, 2004);
2. the area-under-the-receiver operating curve (Marzban, 2004);
3. the log likelihood, and
4. three pseudo- R^2 measures that can be applied with logistic regression.

These measures can be used to compare different models when applied to the same data set. This provides a quantitative measure of how well the modeled predictions fit the observed training data. It is also informative to compare the fit of models to blind tests where the data were not used to develop the model. Thus, we also computed the overall accuracy, positive predictive value, and negative predictive value of each model (with a landslide probability threshold of 50%), as these values can be compared. After computing these measures for models with all possible combinations of landslide susceptibility factors, we also tested different data

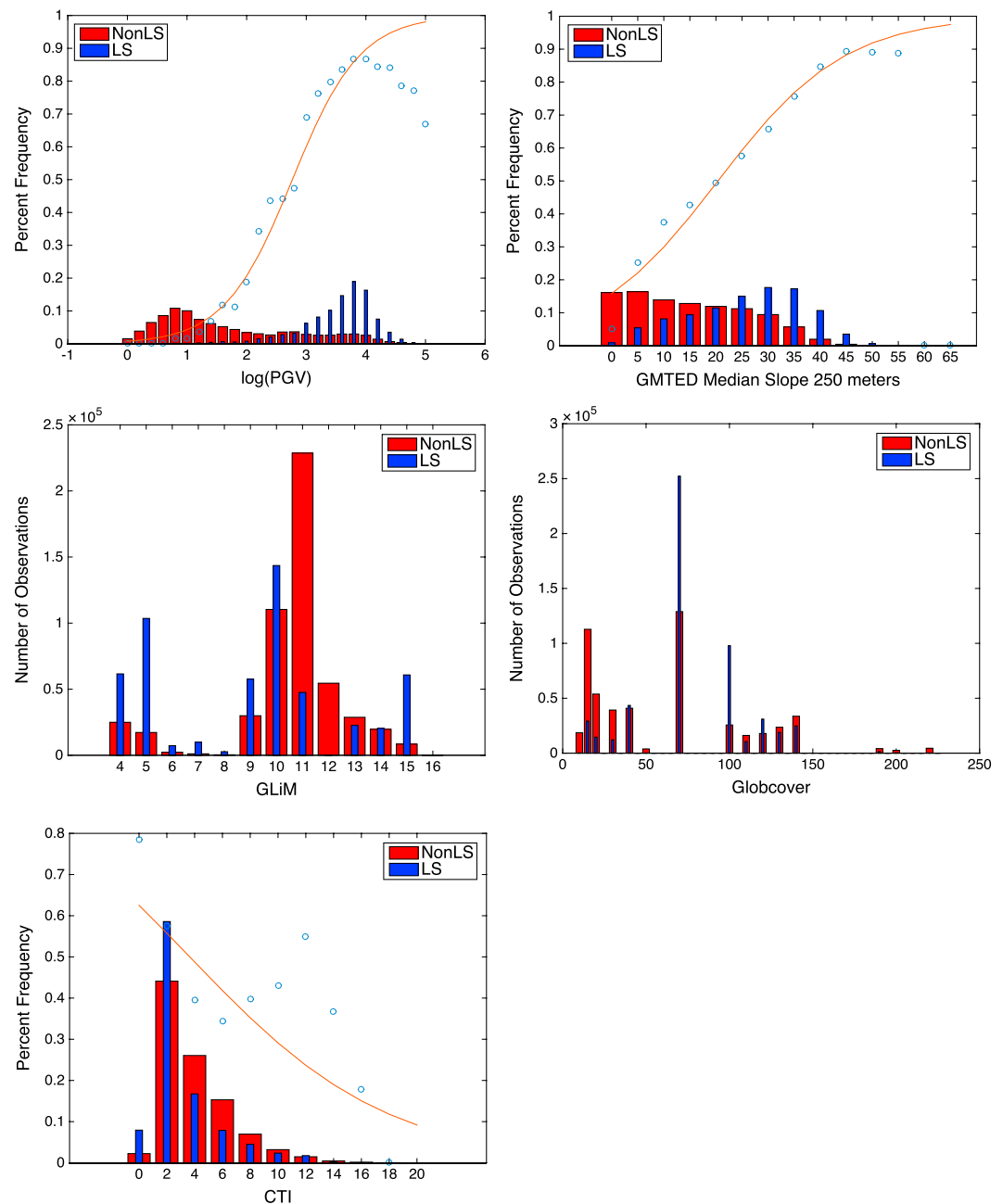


Figure 2. Example histograms of predictor variables ultimately included in the global landslide model. Note the difference in observed values for landslide versus nonlandslide observations (blue and red bars, respectively). Red lines show a logistic regression model between the individual variable and landslide/nonlandslide observations (none shown for nominal data as each value is treated as a separate variable). Blue circles show the fraction of observations that had a landslide in each bar. LS = landslide; NonLS = non-landslide; PGV = peak ground velocity; CTI = compound topographic index; GMTED = Global Multi-resolution Terrain Elevation Data; GLiM = global lithological map.

sets representing each landslide susceptibility factor (when available), common transformations of variables (e.g., natural logarithm and square root), and allowed interaction (multiplication) between highly contributing predictor variables. We chose the *best fit* model by choosing the model that performs best as computed by a majority of these goodness-of-fit measures. We confirm this best fit by evaluating the goodness-of-fit on the remaining measures (that did not agree with the majority) to provide a robust estimate of the performance of the chosen model formula.

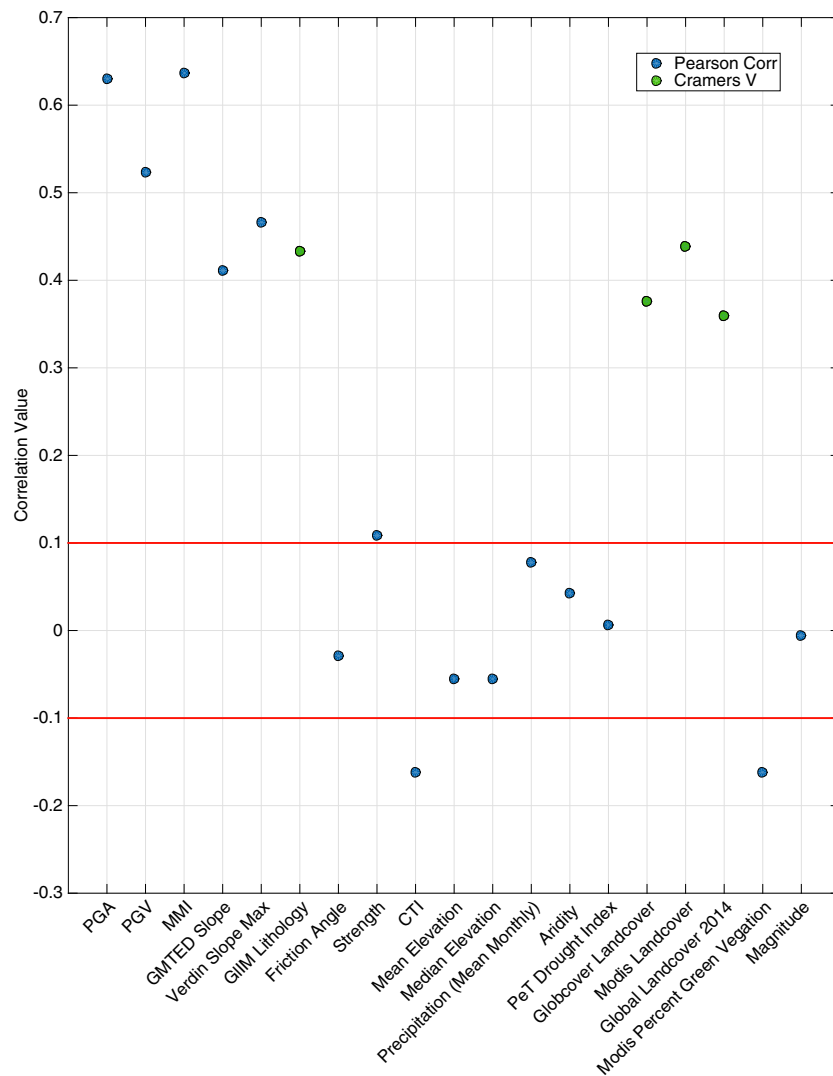


Figure 3. Correlation coefficients computed for individual predictor variables. Blue dots indicate Pearson's correlation coefficient (interval level data); green dots indicate correlation represented by Cramer's V (appropriate for nominal level data). A correlation of less than absolute value of 0.1 (indicated with horizontal red lines) indicates little to no correlation is present between the variable and landslide occurrence, and these variables are removed from the testing process. PGA = peak ground acceleration; PGV = peak ground velocity; MMI = Modified Mercalli Intensity; CTI = compound topographic index; GMTED = Global Multi-resolution Terrain Elevation Data; GLiM = global lithological map.

In order to understand the performance of the model, we compare the observed training landslide observations with modeled results. The landslide observations are binary (yes or no), and the model output is given as a probability; to directly compare the two for testing purposes, we apply a probability threshold of 50%. Therefore, any predicted probability less than 50% is classified as not likely to have a landslide and any probability 50% and over is classified as likely to have a landslide. Note that various probability thresholds can be applied; we choose 50% for consistency with the imposed sampling balance. Consequently, there are four options when comparing the model inputs and outputs: true positives (TP; correctly identified landslides), true negatives (TN; correctly predicted non-landslide areas), false positives (FP; the model predicts a landslide and no landslide was observed), and false negatives (FN; the model predicts no landslide and a landslide was observed). Note that users of the model do not need to classify the outputs in this way for them to be meaningful.

For individual events, we also compute the balanced accuracy using different probability thresholds for classification of the outputs (equation (6)).

$$Acc_{Bal} = \frac{(TPR + TNR)}{2} = \frac{\left[\left(\frac{TP}{TP+FN}\right) + \left(\frac{TN}{FP+TN}\right)\right]}{2}, \quad (6)$$

where TPR is the true positive rate and TNR is the true negative rate. This value represents the ability of the model to correctly identify landslides and nonlandslides equally, regardless of class imbalance. We compute this value to provide a quantitative measure that can be used together with visual analysis to determine how well the modeled predictions are fitting the landslide data shown for individual earthquakes.

3.4. Model Uncertainty Measures

To further understand the landslide probabilities produced by the logistic regression model, we evaluated the uncertainty in these predictions by calculating a standard prediction interval for these predictions. We did so by propagating the standard error measures calculated on individual coefficients in the logistic regression model through to the resulting predicted probabilities:

$$P_I = \text{logit} (P \pm 1.96 \times SE), \quad (7)$$

where P_I = prediction interval, P = predicted probability, SE = standard error, and 1.96 is the value corresponding to the area under 95% of a normal distribution. There is a 95% chance that the actual probability lies within this prediction interval. The standard error on each individual prediction is calculated by using the upper and lower confidence interval bounds for each coefficient in the model to produce an upper and lower bound on the predicted probability. Note that this approach does not take into account data uncertainty but incorporates only the uncertainty in the model calculation (i.e., the quality of the fit of the MLE method). These values thus provide relative constraints on how uncertain the modeled predictions are but likely underestimate the true uncertainty in the probability estimates. A quantitative estimate of the true uncertainty in the predictions should include both the uncertainty in the measurement of the individual variables (aleatory variability) as well as uncertainty associated with model fit (epistemic uncertainty).

4. Results

4.1. Model Selection

We tested 133 different models, with a variety of representations of each input variable, as well as combinations of variables. In order to find the *best fitting* logistic regression model to these landslide data, we solved for models based on combinations of different predictor variables that show a relationship with landslide occurrence. The models include both linear combinations of individual variables, as well as interaction terms (products) of two or more variables. We then computed the goodness-of-fit measures described in section 2.1 to determine how well each model fits the landslide data. Results from the 10 best fitting models (out of the 133 models tested) in this analysis are shown in Table S2 (see Table S3 for full model selection table). Model #1 is chosen as the best fitting model, as it yields the lowest AIC value, the highest value of the three pseudo- R^2 measures (where a value of 1 indicates all of the variability in the landslide data is captured with this model), and a very high area-under-the-receiver operating curve value.

We tested a number of alternate models utilizing PGA as the representation of ground shaking, some of which yielded relatively high-scoring models. Nonetheless, we choose model #1 for two reasons: (1) it yields the best fitting value of all other computed goodness-of-fit measures, and (2) PGA can have large values for moderate earthquakes—high-frequency spikes—that have corresponding PGVs that are relatively low. In addition, PGA saturates for larger magnitude, yet PGV continues to grow (Wald et al., 1999). PGV thus allows for a peak ground motion model that will be applicable for moderate-to-large magnitude earthquakes. Note also that the selected model represents ground shaking by the natural logarithm of the PGV; this transformation allows the data to be distributed more normally than the original value of ground shaking, allowing for a more stable model solution. The final term included in the model is an interaction term between the natural logarithm of PGV and slope. Including this interaction term improves the fit of the model to the data and may act as a proxy for the process of topographic amplification, which is not otherwise incorporated into the ground shaking estimates.

The best fit model has a functional form shown in equation (8):

Table 3
Summary of the Chosen Best Fit model (Model #1)

Variable code	Description	Estimate	Std. error	Lower CI	Upper CI	Odds ratio	Z value	Pr(> z)
(Intercept)	—	−6.30	0.05	−6.40	−6.20	0.00	−124.49	≪0.001
log (pgv)	Ground shaking	1.65	0.01	1.64	1.66	5.21	228.46	≪0.001
gtd75c	Slope	0.06	0.00	0.06	0.06	1.06	59.44	≪0.001
glim75c3	Metamorphics	−1.87	0.03	−1.92	−1.81	0.15	−72.04	≪0.001
glim75c4	No data	−0.66	0.08	−0.82	−0.50	0.52	−8.29	≪0.001
glim75c5	Acid plutonic rocks	−0.78	0.03	−0.84	−0.73	0.46	−29.42	≪0.001
glim75c6	Basic plutonic rocks	−1.88	0.04	−1.95	−1.81	0.15	−52.38	≪0.001
glim75c7	Intermediate plutonic rocks	−1.61	0.04	−1.69	−1.54	0.20	−41.17	≪0.001
glim75c8	Pyroclastics	−1.05	0.04	−1.13	−0.98	0.35	−27.19	≪0.001
glim75c9	Carbonate sedimentary rocks	−0.95	0.03	−1.00	−0.89	0.39	−34.46	≪0.001
glim75c10	Mixed sedimentary rocks	−1.36	0.03	−1.42	−1.31	0.26	−53.16	≪0.001
glim75c11	Siliciclastic sedimentary rocks	−1.92	0.03	−1.97	−1.87	0.15	−74.34	≪0.001
glim75c12	Unconsolidated sediments	−3.22	0.03	−3.28	−3.16	0.04	−107.71	≪0.001
glim75c13	Acid volcanic rocks	−1.54	0.03	−1.59	−1.48	0.22	−56.48	≪0.001
glim75c14	Basic volcanic rocks	−1.50	0.03	−1.55	−1.45	0.22	−56.37	≪0.001
glim75c15	Intermediate volcanic rocks	−0.81	0.03	−0.86	−0.75	0.45	−27.74	≪0.001
cti	Spatially varying wetness	0.03	0.00	0.03	0.04	1.03	27.55	≪0.001
globcover14	Rainfed croplands	0.91	0.03	0.84	0.98	2.48	26.00	≪0.001
globcover20	Mosaic cropland	0.88	0.04	0.81	0.95	2.41	24.65	≪0.001
globcover30	Mosaic vegetation	0.78	0.04	0.71	0.85	2.19	21.46	≪0.001
globcover40	Closed to open broadleaved evergreen or semideciduous forest	0.68	0.03	0.62	0.75	1.98	20.03	≪0.001
globcover50	Closed broadleaved deciduous forest	0.30	0.05	0.21	0.39	1.35	6.59	≪0.001
globcover60	Open broadleaved deciduous forest/woodland	1.77	0.18	1.42	2.12	5.86	9.90	≪0.001
globcover70	Closed needleleaved evergreen forest	1.71	0.03	1.64	1.78	5.53	50.13	≪0.001
globcover90	Open needleleaved deciduous or evergreen forest	−1.26	0.06	−1.38	−1.15	0.28	−22.01	≪0.001
globcover100	Closed to open mixed broadleaved and needleleaved forest	1.50	0.03	1.43	1.57	4.49	43.13	≪0.001
globcover110	Mosaic forest or shrubland/grassland	0.68	0.04	0.61	0.76	1.98	18.58	≪0.001
globcover120	Mosaic grassland/forest or shrubland	1.13	0.04	1.06	1.20	3.10	31.66	≪0.001
globcover130	Closed to open broadleaved or needleleaved, evergreen or deciduous shrubland	0.79	0.04	0.72	0.86	2.20	22.02	≪0.001
globcover140	Closed to open herabaceous vegetation	1.03	0.04	0.96	1.10	2.79	28.85	≪0.001
globcover150	Sparse vegetation	0.54	0.04	0.47	0.61	1.72	14.62	≪0.001
globcover160	Closed to open broadleaved forest regularly flooded	2.34	0.07	2.20	2.48	10.39	33.13	≪0.001
globcover180	Closed to open grassland or woody vegetation on regularly flooded or waterlogged, soil, fresh, brackish, or saline water	1.19	0.48	0.24	2.14	3.29	2.46	0.01
globcover190	Artificial surfaces and associated areas	0.30	0.05	0.20	0.39	1.34	6.09	≪0.001
globcover200	Bare areas	−0.06	0.10	−0.26	0.13	0.94	−0.64	0.52
globcover220	Permanent snow and ice	−0.18	0.04	−0.26	−0.10	0.83	−4.31	≪0.001
globcover230	No data (burnt areas, clouds, ...)	−1.08	0.50	−2.07	−0.09	0.34	−2.14	0.03
log (pgv)*gtd75c	Interaction term	0.01	0.00	0.01	0.01	1.01	27.02	≪0.001

Note. Std. Error = standard error of the calculation in the logistic regression; Lower CI = lower bound on confidence interval; Upper CI = upper bound on confidence interval; Pr(>|z|) = p value.

$$t = \text{Logit}(P) = a + b \times \ln(\text{PGV}) + c \times \text{Slope} + d \times \text{Lithology} + e \times \text{Land Cover} + f \times \text{CTI} + g \times \ln(\text{PGV}) \times \text{Slope}, \quad (8)$$

where a, b, c, d, e, f , and g are coefficients solved for within the regression. Table 3 provides a summary of the chosen model.

5. Model Analysis

5.1. Visualizing Model Results

Figures 4 and 5 show examples of the landslide susceptibility factors for the area surrounding the $M_w 7.9$ 12 May 2008 Wenchuan, China, earthquake and $M_w 7.0$ 12 January 2010 Haiti earthquake, respectively, along with the modeled landslide probabilities and mapped landslides. The balanced accuracy for the Wenchuan, China, map is 94.6% (when a 50% probability threshold is imposed; i.e., every probability of

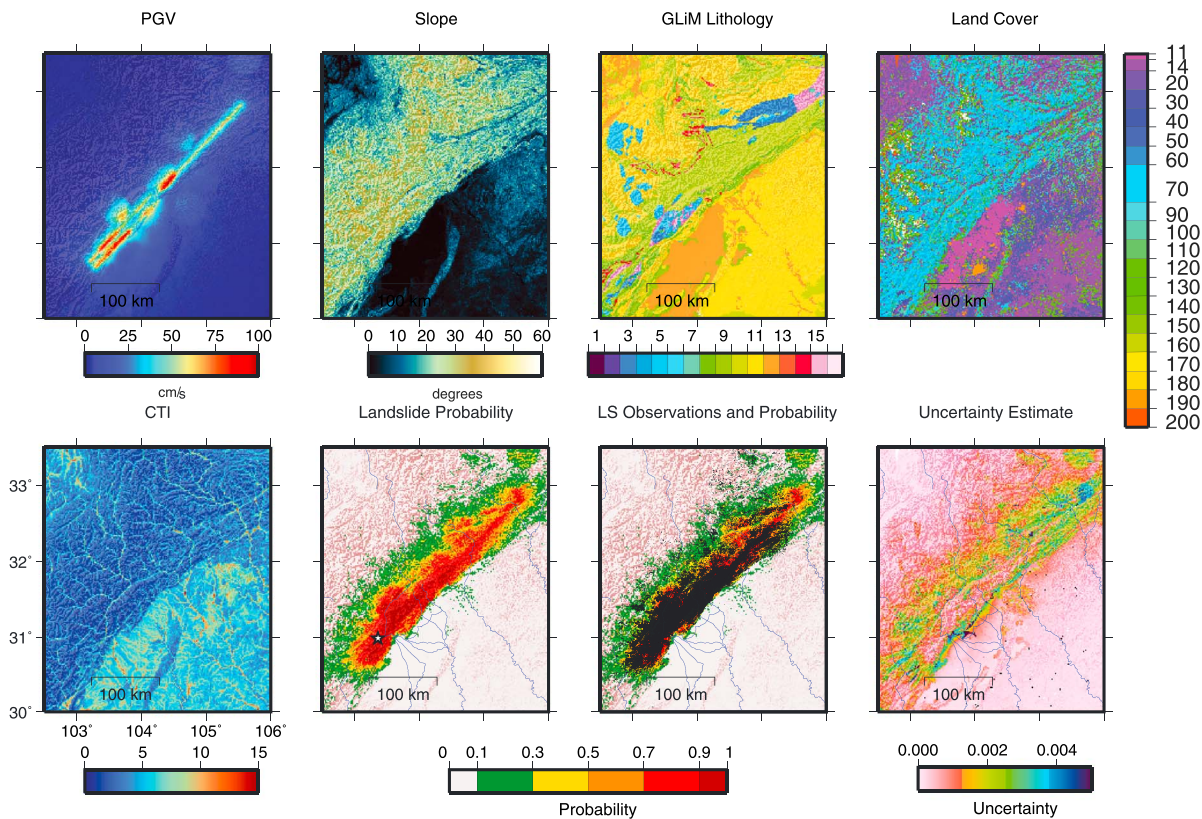


Figure 4. Predictor variables and predicted landslide probabilities for the area surrounding the M_w 7.9 12 May 2008 Wenchuan, China, earthquake. Location of mainshock epicenter is shown by a star. Black dots indicate landslides used for training the model (Xu, Xu, Yao, et al., 2014). Blue lines indicate rivers. Uncertainty estimates are shown as the range between highest and lowest predicted probabilities that fit the logistic regression well. PGV = peak ground velocity; CTI = compound topographic index; LS = landslide; GLiM = global lithological map.

50% and above is classified as a landslide; see section 3.3 for details of how this is computed). The balanced accuracy of the Haiti map is 93.7% at a 50% probability threshold. Figure 6 illustrates examples of forward estimates from our best fitting model formula as applied to the M_w 7.7 20 September 1999 Chi-Chi, Taiwan, and M_w 6.9 17 January 1995 Kobe, Japan, earthquakes, respectively, with the landslide training data superimposed on the landslide probabilities (panels b and f) and uncertainty measures (panels c and g, computed as described in section 3.4). These maps give a good qualitative indication that the model is resulting in high landslide probabilities where landslides have occurred in the training data. The balanced accuracy value also reflects this, in the Chi-Chi, Taiwan, map with 92.5% accuracy at a 50% probability threshold, while the Kobe, Japan, map has an accuracy of 93.1%. The predicted map of landsliding due to the Kobe, Japan, earthquake shows an area of predicted high probability in the northeastern portion of the map, with no mapped landslides. This area lies outside of the area that was surveyed for landslides, and therefore, it is unknown whether or not landslides occurred in this region. These figures also show the prediction interval for landslide probability of these individual training events (as described in section 3.4).

To examine the predicted landslide probabilities in more detail, we zoom into a smaller region in the south central portion of the Chi-Chi, Taiwan, map (Figure 6). Upon visual inspection of this smaller area, we see that the regions with the highest landslide probabilities (>90%) are also the areas of the map with highest mapped landslide density. We can also see features (e.g., ridgelines in the north central portion of the map) that show a distinct pattern of landsliding, with high landslide probabilities following the shape of the feature closely. Maps showing probability results in comparison to the remaining training inventories can be found in Figure S3.

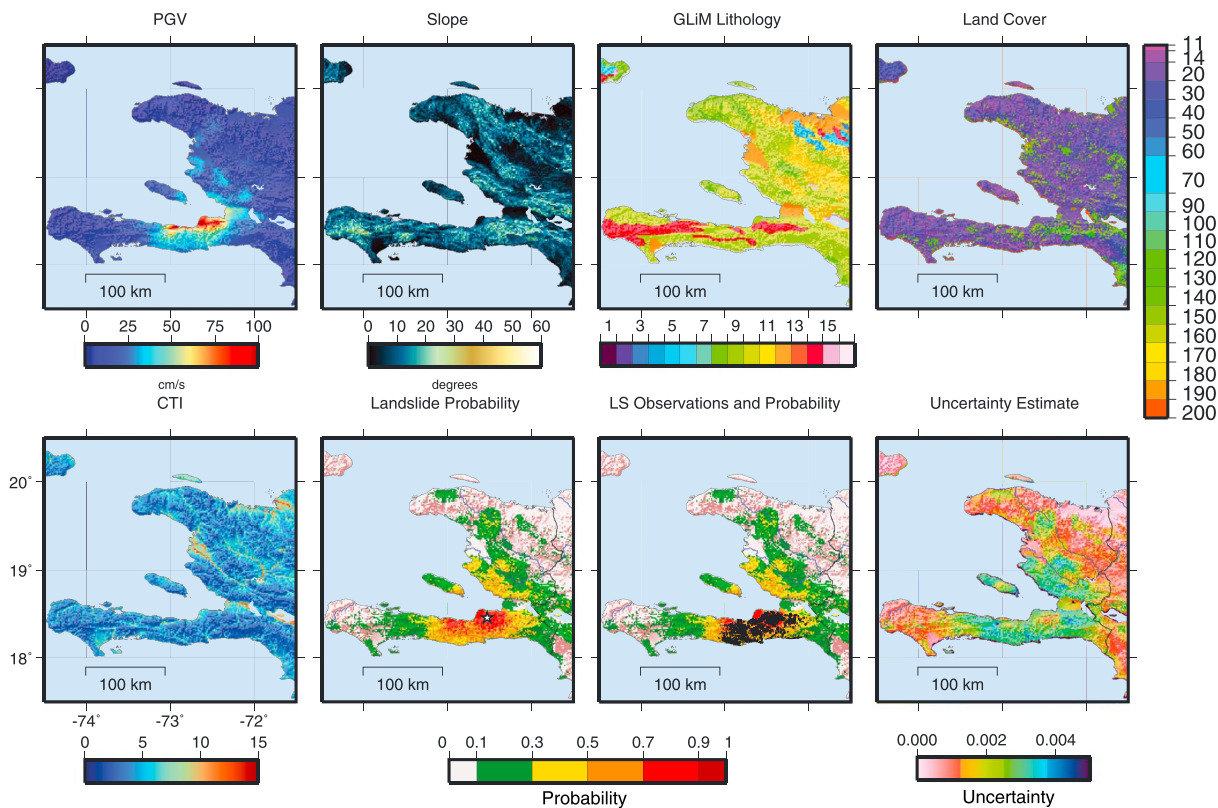


Figure 5. Predictor variables and predicted landslide probabilities for the area surrounding the M_w 7.0 12 January 2010 Haiti earthquake. Location of mainshock epicenter is shown by a star. Black dots indicate landslides used for training the model (Harp et al., 2016). PGV = peak ground velocity; CTI = compound topographic index; LS = landslide; GLiM = global lithological map.

5.2. Visualizing Modeled Data in Parameter Space

While qualitative visualizations provide evidence of the model fitting the data, they provide little information about the predictor variables used within the model. In order to describe the global database as a whole, we plotted the landslide and nonlandslide observations in the parameter space of pairs of individual predictor variables, as shown in Figure 7. This provides a visualization of both the range of observations and pattern of landsliding for individual predictor variables. These plots then show visual patterns within the data where each individual box on these plots shows the portion of observations that experienced those particular parameter values that experienced a landslide. The most apparent pattern shows an increase in landslide density per number of observations as PGV and slope increase. There is less dependence on CTI, but the data do show a small decrease in the proportion of landslide density per number of observations as CTI increases (note that CTI is an additional descriptor of topographic form and therefore has a built-in dependence on slope, which is already taken into account in the model). On top of these observations, we show contour lines of the modeled landslide probabilities, confirming that the modeled predictions follow the pattern of landsliding shown in the data. Figures 7c and 7d then show the observations of the global database in terms of slope on the abscissa, with land cover and lithology classes, respectively, on the ordinate. Figure 7c shows clearly that certain land cover types experience a higher proportion of landsliding, such as mixed forests or needle-leaved evergreen forests. Figure 7d shows that certain lithologies experience landsliding at different slope angles; for example, plutonic and sedimentary rocks tend to fail at higher slopes, while pyroclastic rocks tend to fail more at gentler slopes. We also see that a majority of lithologies that experience more landsliding have observations within the database at higher slopes, while lithologies that are only represented at low slopes (such as unconsolidated sediments) exhibit very little landsliding at the slope angles for which we have data. These plots can aid in interpreting the model coefficients, as these variables may be codependent on slope.

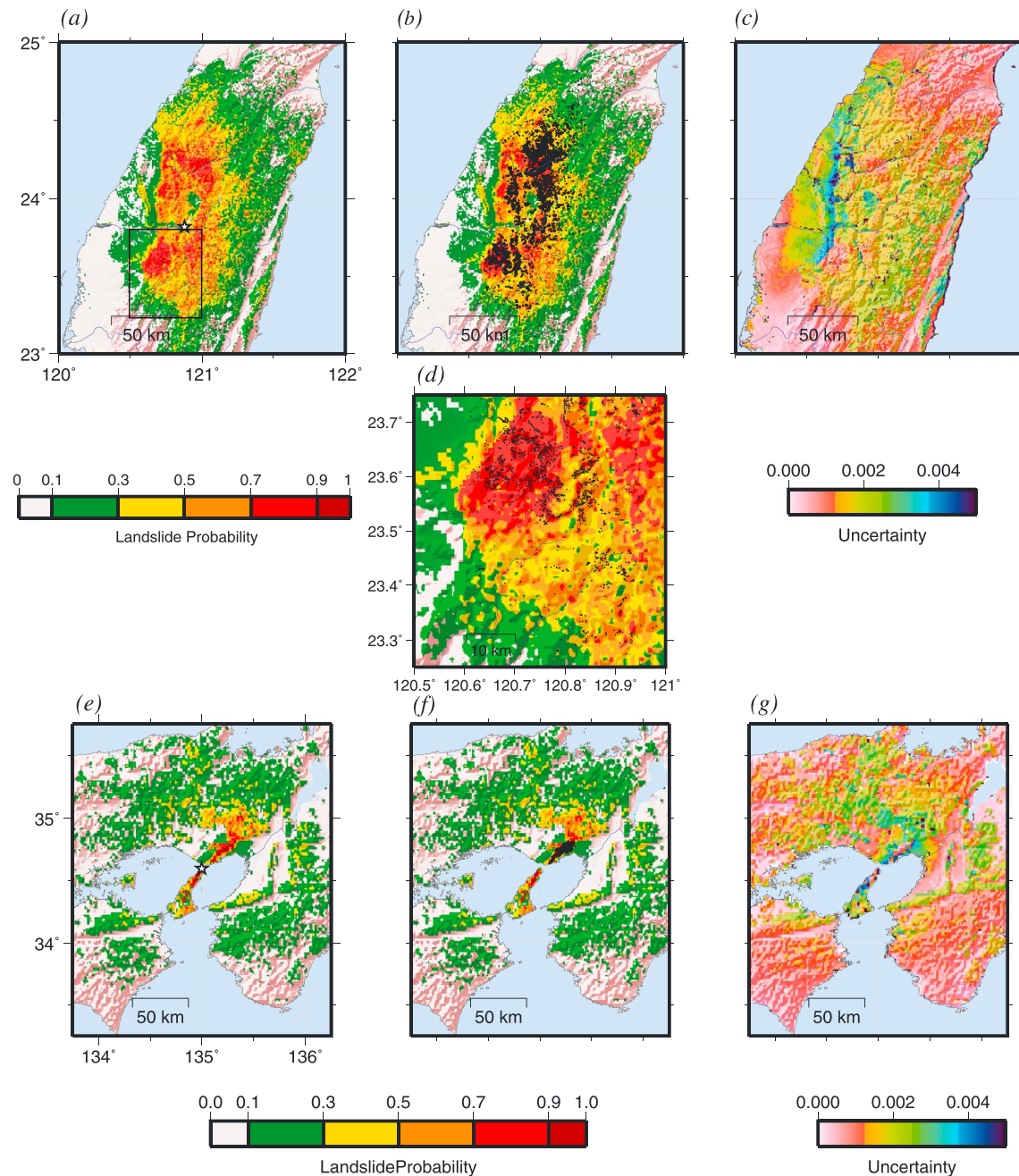


Figure 6. (a) Landslide probabilities for the $M_w 7.7$ 20 September 1999 Chi-Chi, Taiwan, earthquake. (b) Landslide probabilities underneath black dots that indicate landslides used for training the model (Liao & Lee, 2000). (c) Uncertainty estimates out of the logistic regression model. (d) Landslide probabilities underneath black dots that indicate landslides for a smaller region of Taiwan (extent shown by the black box in part a). (e) Landslide probabilities for the $M_w 6.9$ 17 January 1995 Kobe, Japan, earthquake. (f) Landslide probabilities underneath black dots that indicate landslides used for training the model. (g) Uncertainty estimates out of the logistic regression model. Location of mainshock epicenter is shown in parts a and e by star.

5.3. Interpreting Odds Ratios of Independent Variables

Table 3 also provides the odds ratios for each variable included in the chosen model. Unlike the probabilities estimated by the logistic regression model, the odds ratio for landslide occurrence (defined as $P_{\text{landsliding}}/P_{\text{not landslide}}$) has a linear response with each independent variable, thus showing how much the probability of landsliding increases for every unit increase in each individual variable. For example, for every unit increase in CTI, the odds of landslide occurrence increase by 1.03. These values can

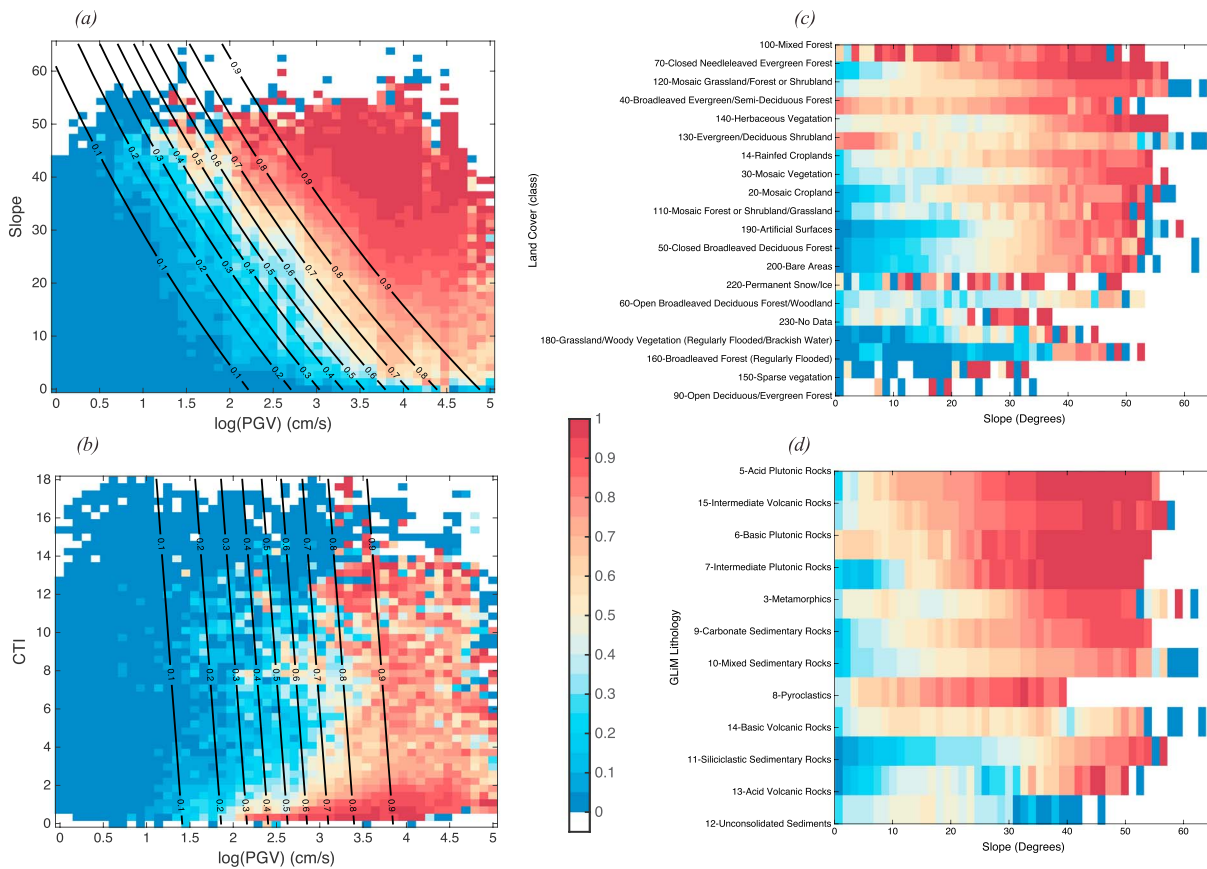


Figure 7. Visual representation of the global database, plotted in terms of (a) PGV and slope, (b) PGV and CTI, (c) slope and land cover, and (d) slope and lithology. Colors represent the portion of observations within each grid cell that had a landslide, from no landslides (blue) to all landslides (red). Contours on plots (a) and (b) show the model probability in terms of the chosen parameters. PGV = peak ground velocity; GLIM = global lithological map; CTI = compound topographic index.

explain how sensitive the overall likelihood of landsliding is to each individual independent variable but are dependent on the units of each variable and therefore cannot be compared directly. Interpreting these values shows that an increase in the natural log of PGV by 1 cm/s increases the odds of landsliding by 5.21, and an increase in slope by 1° increases the odds of landsliding by about 1.06. This analysis also shows that the presence of certain land cover or rock type class correlates with landsliding. The four highest positive correlations from land cover type are broadleaved forests that are regularly flooded with brackish water (likely due to a small sample size of this rare land cover classification), open broadleaved deciduous forest/woodland, closed needle-leaved evergreen forest, and mixed broadleaved/needle-leaved forest. The four highest associations with lithology are areas with no lithology data present (potentially because it is more difficult to map steep and rugged areas where landslides are more likely to occur), acidic (felsic) plutonic rocks, intermediate volcanic rocks, and carbonate rocks. While the presence of these individual lithology or land cover types increase the odds of landsliding, the outcome may be codependent on slope, which inherently may be controlled by lithology and land cover type. For example, unconsolidated sediments are shown to correspond with landsliding probability the least, but a majority of unconsolidated sediments are in areas of low slope less than 45° (Figure 7d) and therefore are less likely to landslide in general.

5.4. Statistical Measures of Accuracy

Ultimately, the success of this model depends on how accurately it predicts the sites of seismically induced landslides. Figure 8a shows the results of the classification scheme (described in section 3.3) when applied to the final landslide model. This histogram shows the distribution of predicted landslide probabilities, as well as

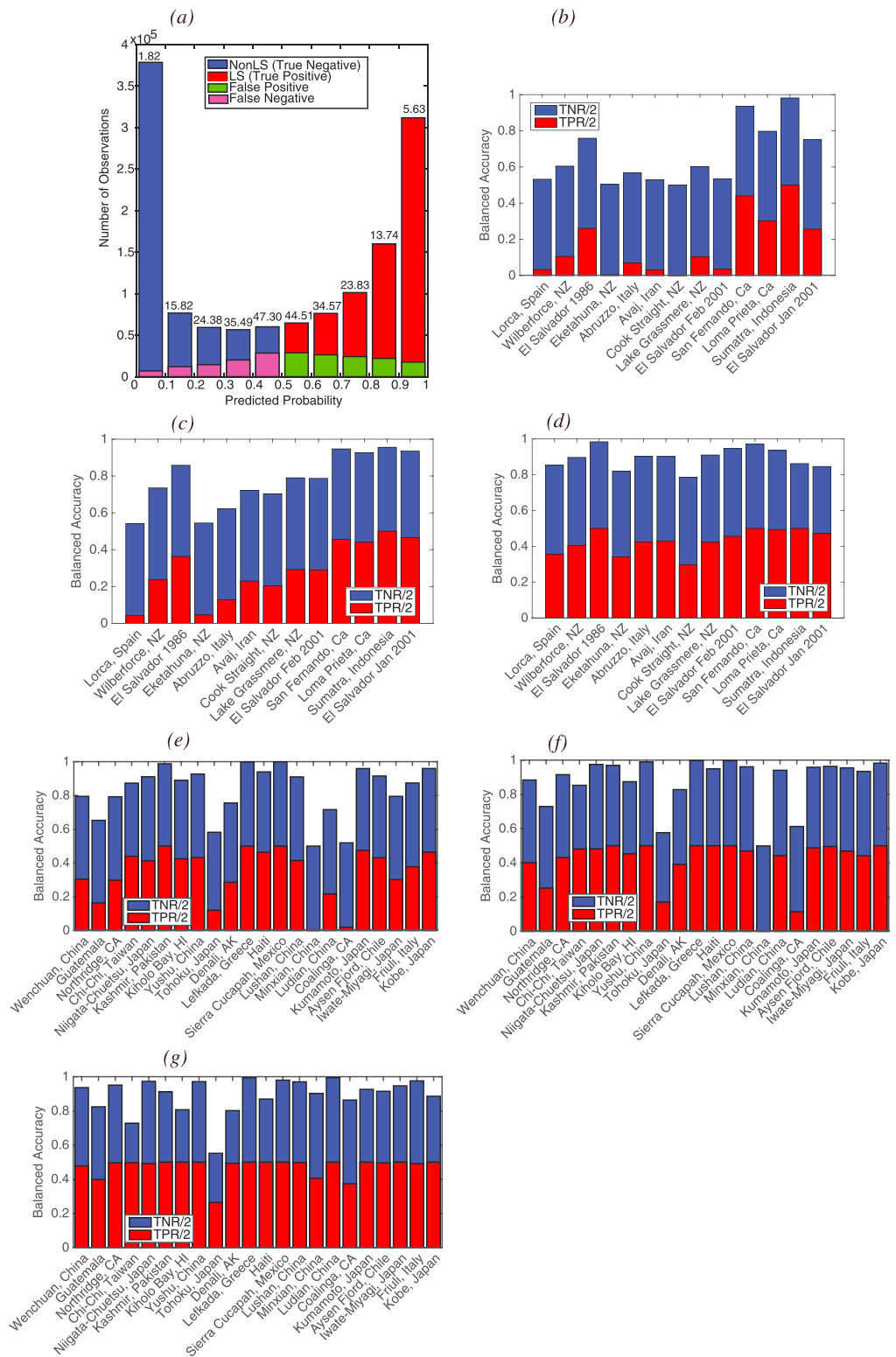


Figure 8. Histogram of predicted probabilities from the (a) global landslide training data set. Applying the model to blind test events and imposing a (b) 50%, (c) 30%, and (d) 10% landslide probability threshold. Bar graph of error measures as applied to each earthquake in the leave-one-event-out cross validation analysis. Models are solved for using all data except one event, and applying that model to that event to see how well it fits the data by imposing a (e) 50%, (f) 30%, and (g) 10% landslide probability threshold. Blue and red indicate *true* predictions. TNR = true negative rate; TPR = true positive rate; LS = landslide; NonLS = nonlandslide.

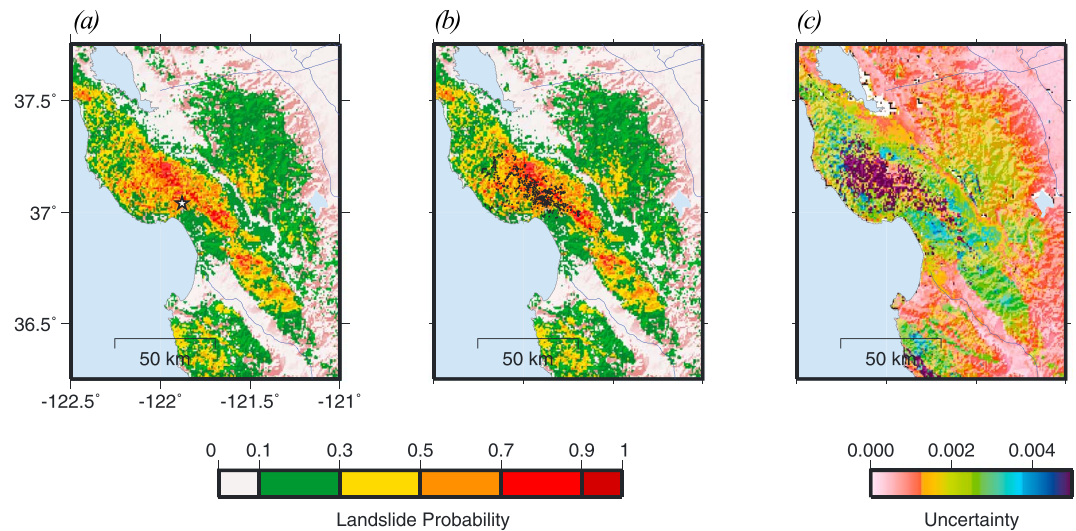


Figure 9. Blind validation showing (a) predicted landslide probabilities for the Mw6.9 27 June 1989 Loma Prieta, California, earthquake). Location of mainshock epicenter is shown by a star. (b) Mapped landslides are shown as black dots (Keefer & Manson, 1998) and were not used to train the model. These are plotted on top of predicted probabilities for comparison. Note the correlation of high landslide probabilities and mapped landslides, indicating good model performance. (c) Uncertainty estimates out of the logistic regression model.

the percentage of landslide observations correctly and incorrectly identified by the global model with a probability threshold of 50%. The percentages of false positives and false negatives then represent an estimate of the inability of the model to correctly identify landslide observations it was trained with, while the percentages of true positives and negatives represent the ability of the model to correctly identify landslide observations. Note that the shape of this histogram reflects the nonlinear shape of the logistic function, predicting more probabilities toward 0 and 1 with fewer values in between. Overall, the low percentages of false positives and false negatives demonstrate a good fit to the landslide data. As expected, the number of *correct* and *incorrect* predictions is approximately equal as the predicted probability approaches 50%.

After computing the model equation, we also determine if multicollinearity exists between predictor variables by computing the variance inflation factor (VIF; Chatterjee & Price, 1977). Any calculated VIF of 10 or above has been shown to exhibit large multicollinearity, indicating that one or more of the predictor variables corresponds with another predictor variable in the model. By performing this test, we show whether the predictor variables are independent of one another, and therefore, each is providing new information to the model. Were a VIF very high, it would indicate that a variable should be removed from the model equation, as two or more of the predictor variables are dependent on one another. The results show low VIF values for all of the variables within the model except for PGV and slope, as these two variables are also represented in the model a second time within the interaction term. To determine whether or not PGV and slope exhibit multicollinearity with one another, we calculate the VIF value for a model without the interaction term and find the VIF values are less than 4, indicating low multicollinearity between all of the primary predictor variables.

5.5. Testing for Robust Applicability

5.5.1. Performance of Blind Test Events

Our goal is to predict where landslides will occur based on known information that is readily available and real-time data describing the amount of shaking for a specific earthquake. In order to simulate this scenario of application (i.e., events for which we do not a priori know the distribution of landsliding), we apply the model to earthquakes that were not used to train the model and compare the predicted probabilities to these known landslide distributions, i.e., a blind test of the model.

Figure 9 shows the application of the model to the M_w 6.9 27 June 1989 Loma Prieta, California, earthquake, an event with available landslide data that were not used in calculating the model. Since this is an incomplete inventory, it is not possible to compare the modeled probabilities with locations outside of the mapped

landslide area. Upon visual inspection of the predictions, we see that landslide observations from the earthquake match well with predicted high probabilities, yet the model predicts potential landsliding in a large area outside of the mapped landslide area. The balanced accuracy of 85% (at a 50% probability threshold) supports this conclusion. Maps showing probability results in comparison to the remaining blind test inventories can be found in Figure S4.

To quantitatively determine how well the model performs on blind test events, we calculate the TPR, TNR, and balanced accuracy of the remaining earthquakes that were not included in the training dataset. Upon doing so, we compare the modeled predictions to the actual landslide distributions statistically, as seen in Figure 8b (50% probability threshold), Figure 8c (30% probability threshold), and Figure 8d (10% probability threshold). Recall that this test compares modeled landslide probabilities to the lower scoring landslide inventories; many of these inventories are not complete but can be compared to modeled results for qualitative validation. The results indicate that all of the events have a balanced accuracy of greater than 50%. This indicates that the modeled landslide probabilities for all of the events accurately match greater than 50% of the pixels. When applying a lower (30%) probability threshold, the average balanced accuracy value across all of the events is 77.5%. As the probability threshold used to classify landslides is lowered to 30% (Figure 8c) and 10% (Figure 8d), the predictive ability of the model as applied to these events increases. The events with the lowest magnitudes show the most change in TPR as the probability threshold is lowered, indicating accurate classification of more landslides. This is likely because these test events are smaller earthquakes with lower modeled landslide probabilities. When using a 10% probability threshold for classification, all of these events have a balanced accuracy of at least 80%, indicating a high percentage of correctly identified pixels.

5.5.2. Leave-One-Event-Out Cross Validation

To simulate conditions similar to how the model would run in near real time, and to determine how well the model would fit future earthquakes, we applied a variation of a standard leave-one-out cross validation method (Hawkins et al., 2003). For this test, we use only the high-quality training data set. However, we leave one event out of the model training set (to later be used for testing), train the model based on the other available earthquakes, and apply this model to the remaining *left-out* earthquake that was not used to train the model. We do this for each earthquake in the original global data set and test how well the models perform when applied to each individual earthquake. We again calculate the TPR, TNR, and balanced accuracy of the “left-out” data. We compare the modeled predictions to the actual landslide distribution statistically, as seen in Figure 8e (50% probability threshold), Figure 8f (30% probability threshold), and Figure 8g (10% probability threshold). This analysis shows that certain events perform better than others at correctly identifying the pattern of landsliding that occurred. In this figure, red and blue indicate *true* predictions (positive and negative, respectively). When applying a 50% probability threshold for classification, the results indicate that all but five of the events have a true positive rate of greater than 50%. All but one event have a balanced accuracy of greater than 50%, indicating that the modeled landslide probabilities for all of the events accurately match greater than 50% of the pixels. When applying a lower (30%) probability threshold (Figure 8f), the true positive rate for a number of the events increases, indicating a number of landslide observations corresponding to predicted landslide probabilities between 30 and 50%. As the probability threshold used to classify landslides is lowered to 10% (Figure 8g), the predictive ability of the model as applied to these events increases even further. The events with the lowest and highest magnitudes show a substantial change in TPR, indicating accurate classification of more landslides. When using a 10% probability threshold for classification, all of these events have a balanced accuracy of at least 70%, indicating a high percentage of correctly identified pixels, save for the Tohoku earthquake. We see that the $M_w 7.9$ 12 May 2008 Wenchuan, China, earthquake has a high true rate, indicating that it performs well, even when left out. The $M_w 9.0$ 11 March 2011 Tohoku, Japan, earthquake has the lowest true rates, indicating that it performs the worst. Note that the Tohoku, Japan, event was the largest earthquake included in the training database; because it performs the worst when left out, this may indicate that a magnitude term may be needed within the model to accurately model larger events and account for duration of shaking. The variability in these results may indicate that future models should incorporate uncertainty estimates from the ground motion models. The consistent results of this simulation analysis across a majority of the earthquakes indicate that the model can be applied robustly to new earthquakes as they occur, with an expectation of occasional outlying events that will only be partially matched by the global model.

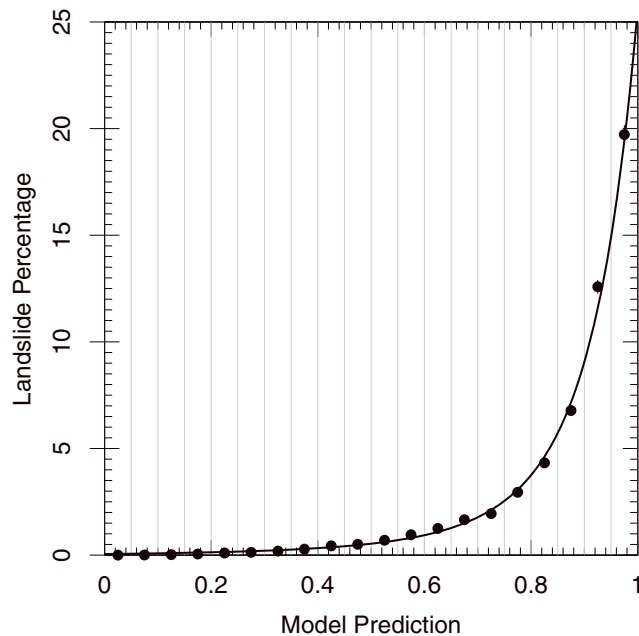


Figure 10. Relationship between predicted model probabilities and frequency of landslide occurrence used to represent the actual probability of a landslide occurring (i.e., areal coverage of landsliding expected in a given grid cell).

5.5.3. Converting Probabilities to Areal Percentages

As described previously, we have used a sampling balance of 1:1 for landslide occurrences to nonoccurrences. Including inventories for which the true ratio of occurrences to nonoccurrences is unknown forces the selection of an arbitrary sample class balance. Oommen et al. (2011) demonstrated that if the class balance of the observations is not the same as the true class balance then the resulting probabilistic model will be severely biased. We address this problem by developing a modification factor from only the comprehensive landslide inventories in order to remove this bias. We sample the inventories on a mesh of uniformly spaced points, classified as occurrences if they fall within polygons of mapped landslides. This sampling method is unbiased with respect to the true class balance. We also tabulate the probabilities of the model (equation (9)) at all of these points. Figure 10 plots the percentage of landslide occurrences within 0.05 width bins of the predicted probabilities. We fit the following curve to these data

$$L_P(P) = e^{(a+b \times P + c \times P^2 + d \times P^3)}, \quad (9)$$

where $a = -7.592$, $b = 5.237$, $c = -3.042$, and $d = 4.035$. This equation can be used to correct the predicted probabilities (i.e., the relative hazard, P) to represent the frequency of landslide occurrence, which can be interpreted as the portion of each cell that is expected to have landslide occurrence (i.e., areal coverage, L_P).

6. Discussion

These statistical tests demonstrate that our approach could be effectively used, with relatively high confidence, to provide near-real-time (e.g., within minutes) estimates of the spatial distribution of landslide probabilities. In principle, this model can be applied after the occurrence of any earthquake on the globe for which a *ShakeMap* is generated. While this relationship is likely to vary with location, our model provides a reliable estimate of the typical relationship between landslide occurrence and geologic and geographic conditions in an area. Given the framework of the model, it can be applied immediately after an earthquake is located and a *ShakeMap* is generated (typically automatically), as this is the only triggering information needed for estimating landslide probabilities for individual earthquakes. Landslide predictions can be produced within minutes of *ShakeMap* production (Allstadt, Thompson, Hearne, Wald, et al., 2018), and estimates will improve with time as more sophisticated versions of the *ShakeMap* are produced, incorporating more information such as ground motion observations, intensity data, and finite fault models (Worden & Wald, 2016). Different *ShakeMap* versions will produce substantially different landslide probability distributions. This was investigated by Allstadt, Jibson, et al. (2018), who applied three globally applicable landslide models to the $M_w 7.8$ 14 November 2016 Kaikoura, New Zealand, earthquake. Their results showed that the estimated landslide distribution is highly dependent on the version of the *ShakeMap* used as an input into the landslide model, which indicates the strong dependence of landslide distribution on strong ground motion. Their results also indicated that once the point-source earthquake model was replaced with an approximate rupture extent, the model successfully identified the approximate area of highest landslide hazard. Given this strong correlation, and the uncertainties inherent in predicting strong ground motions, this limitation should be communicated clearly to users of the landslide model.

This model is built similarly to the model presented in Nowicki et al. (2014). It has been updated with additional data and therefore is more representative of landslide distribution and occurrence. This can be shown by comparing the results of the Nowicki et al. (2014) model to the updated model (Figure 11). Visually, the output from the updated model is smoother (less variation within the area of mapped landslides) and captures smaller clusters of landslide occurrence than the 2014 model. While these outputs cannot be compared statistically in a direct way, due to differences in model resolution and the training method used for development, we can compare the model formulas from both using the current training data and

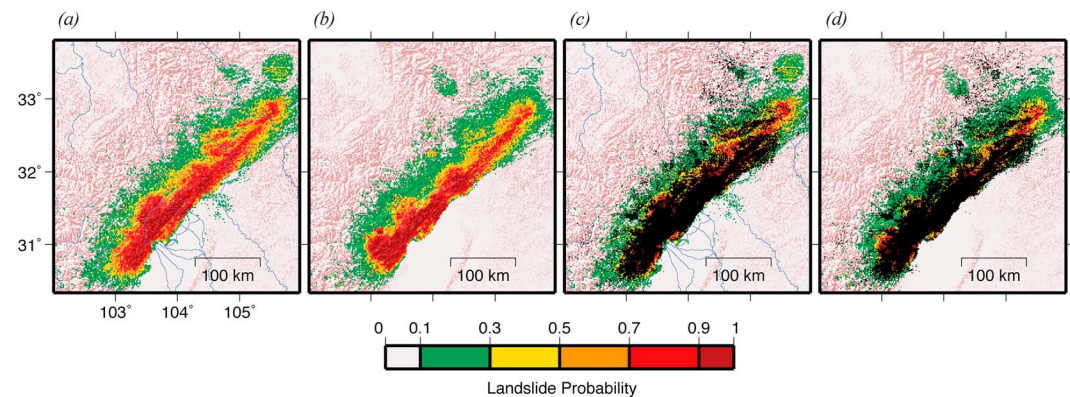


Figure 11. Comparison of results of the proposed model (panels a and c) with results of the Nowicki et al. (2014) model (panels b and d), both applied to the Wenchuan, China (2008) earthquake. Panels c and d show the probability distribution from each of the models, with mapped landslide data from Xu, Xu, Yao, et al. (2014) plotted overtop.

application of the model. Upon doing so, the AIC values are 999,235 (2014 model) and 905,715 (updated model); this shows that the updated model fits the landslide training data more closely as a whole (lower AIC value) than the 2014 model.

At this stage, the model has been tested with available historical data and has been incorporated for preproduction testing into the USGS Ground Failure earthquake product, a new addition to the suite of USGS near-real-time earthquake products that produces estimates of liquefaction and landslide hazard (Allstadt, Thompson, Hearne, Wald, et al., 2018; Allstadt, Thompson, Hearne, & Biegel, 2018). Testing to determine the best way to display the output landslide probabilities of this model, as well as combining with other similar models will enable a robust estimate of landslide hazard due to individual earthquakes (Allstadt et al., 2016; Allstadt, Thompson, Hearne, Wald, et al., 2018). Overlaying roadways and cities with probability of landsliding will make it such that valuable infrastructure can be highlighted and displayed within the model output.

Given the potential for fatalities due to EQILs, care will be taken in developing, displaying and alerting these results to the public and aid agencies. As a whole, the model performs well when evaluated over the domain of an entire event, but clearly, individual pixels will predict probabilities that underestimate or overestimate the landslide risk. Given this reality, care must be taken in choosing an appropriate way to represent these estimates and their inherent uncertainties. These results give information regarding where more detailed analysis could be focused and can inform aid agencies of regions that are more likely to have had landslides occur. These regional-scale calculations show the overall extent of the expected landslide hazard, which may be helpful to aid agencies for evaluating response strategies soon after an earthquake's occurrence.

At present, the model can be applied for any earthquake that has already occurred for which a *ShakeMap* or similar shaking grid is available; it can also be applied to past and hypothetical future earthquakes including the *ShakeMap* archive and its catalogs of scenarios (e.g., the USGS *ShakeMap* scenario archive available at <https://earthquake.usgs.gov/scenarios/catalog/bssc2014/>). The *ShakeMap* shaking scenarios describe the shaking from hypothesized earthquakes of a given size and location; the *ShakeMap* grids can easily be run through the landslide model, with the output as a landslide probability map for the individual scenario earthquake. Given this information, city planners, government officials, and citizens can learn about their potential exposure to future earthquake-triggered landslides before a devastating event occurs. The results can then be incorporated into earthquake hazard preparedness drills or infrastructure vulnerability assessments.

Because this statistical model assumes a stationary relationship between the predictor variables (ground shaking, slope, lithology, ground wetness, and land cover) and landslide occurrence, there are a number of limitations inherent in the model. Although it can be applied anywhere, this relationship may not fit as accurately as a region-specific model, or a model trained to individual climate or tectonic regimes (although data for training local regional models are much more limited). While this global relationship provides a good fit to the data overall, further work should continue to refine models that can be applied accurately for more complex distributions of landsliding that may be impacted by location or climate type.

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Recognizing that the current model may not accurately represent every region of the globe or climate/tectonic regime, efforts to collect additional landslide inventories to train region-specific models should continue. Additional predictor variables may also be tested and incorporated into the model, including more accurate proxies for precipitation prior to an earthquake to determine whether precipitation can be incorporated into the model as a proxy for temporal wetness. Where available, soil thickness data could also be tested to determine the relationship between landslide occurrence and near-surface weathered materials. We also note that this model does not test or include any representation of how landscapes have been altered due to human use. Future testing to include this information could test data that represent modification by roads (e.g., distance to a road) and other substantial land-use change (e.g., deforestation and construction).

Further work could also move toward applying this type of model at the highest resolution possible, as landslide susceptibility is highly associated with local variability in slope and soil/rock conditions. Incorporation of higher-resolution data will require additional computational resources, yet may improve the ability of the model substantially. As more accurate and higher resolution datasets become available, such as topographic slope, CTI, and land cover data, they can be used to train the model.

While we present formal estimates of uncertainty for model predictions, those estimates do not take into account the uncertainty associated with the individual input parameters. While this assessment is beyond the scope of the current study, the ultimate goal of this modeling effort should include provision of realistic estimates of the true uncertainty associated with the output landslide probabilities. Currently, there is a value for uncertainty of ground shaking values, but the remaining predictor variables do not have uncertainty measurements associated with them. Because of this, Bayesian statistics could be used to build synthetic uncertainty values to give an estimate of spatial and event-specific variability in landslide probability values.

7. Conclusions

Data from 23 earthquakes of different sizes and geographic and tectonic settings were combined into a global database of seismically induced landslide occurrence. These data show that a relationship exists between landslide occurrence and a number of predictor variables (as shown in equation (7)), providing a prediction of landslide probability based on globally available predictor variables. Visual and statistical inspection of the model show that the selected model performs well on both training data and with data that were not used to train the model, supporting the application of the model on future earthquakes. As a whole, the model performs well when evaluated over the domain of an entire event. However, this statistical model will always show probabilities that either underestimate or overestimate the actual landslide risk in individual pixels. These regional estimates of landslide hazard can show the overall expected extent of hazard due to landsliding, which may be helpful for evaluating response strategies immediately following an earthquake's occurrence. With continued testing and validation, the model can be incorporated into the USGS Ground Failure earthquake product, as well as the USGS PAGER system and other loss-estimation systems for near-real-time estimation of landslide probabilities triggered by individual earthquakes within minutes of their occurrence.

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